

DSC 335, Handout 01: The Big Bayesian Picture (*Chapter 1*)

LEARNING OBJECTIVES.

- *Compare and contrast the Bayesian and frequentist statistical philosophy.*
- *Describe how the Bayesian statistical philosophy incorporates the ideas of prior and posterior information.*

Probability Quiz: For a moment, forget everything that you have been taught in probability and statistics classes and answer the following questions based on gut instinct:

1. When flipping a fair coin, we say that “the probability of flipping Heads is 0.5.” How do you interpret this probability?
 - (a) If I flip this coin over and over, roughly 50% will be Heads.
 - (b) Heads and Tails are equally plausible.
 - (c) Both a and b make sense.
2. An election is coming up and a pollster claims that candidate A has a 0.9 probability of winning. How do you interpret this probability?
 - (a) If we observe the election over and over, candidate A will win roughly 90
 - (b) Candidate A is much more likely to win than to lose.
 - (c) The pollster’s calculation is wrong. Candidate A will either win or lose, thus their probability of winning can only be 0 or 1.
3. Consider two claims. (1) Zuofu claims that he can predict the outcome of a coin flip. To test his claim, you flip a fair coin 10 times and he correctly predicts all 10. (2) Kavya claims that she can distinguish natural and artificial sweeteners. To test her claim, you give her 10 sweetener samples and she correctly identifies each. In light of these experiments, what do you conclude?
 - (a) You’re more confident in Kavya’s claim than Zuofu’s claim.
 - (b) The evidence supporting Zuofu’s claim is just as strong as the evidence supporting Kavya’s claim.
4. Suppose that during a recent doctor’s visit, you tested positive for a very rare disease. If you only get to ask the doctor one question, which would it be?
 - (a) What’s the chance that I actually have the disease?
 - (b) If in fact I don’t have the disease, what’s the chance that I would’ve gotten this positive test result?

The Meaning of Probability

Asking questions

Bayesian Knowledge Building Diagram

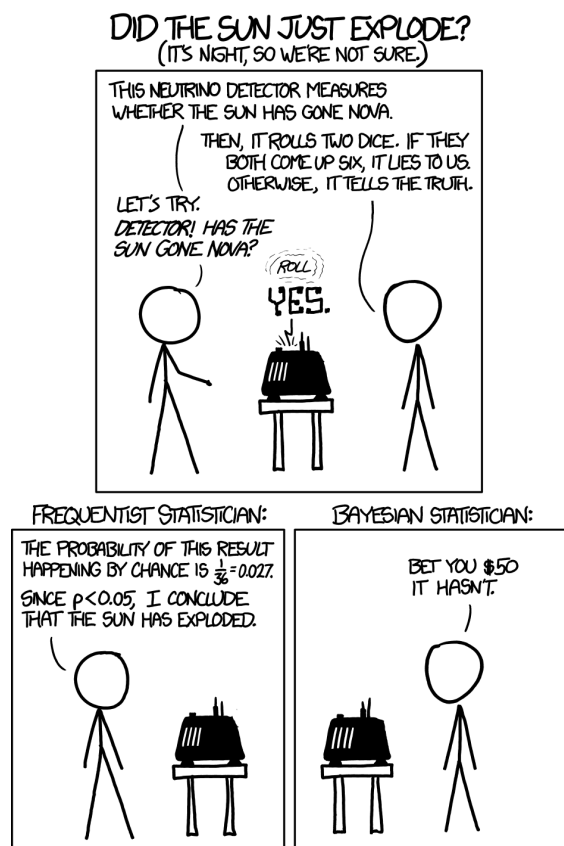
Frequentist Knowledge Building Diagram

The Bayesian Balancing Act

Example 1. Consider the scenario from the probability quiz, question 4. The table below gives disease status and outcome status for 100 people. What questions could a Bayesian or frequentist ask from the data?

	test positive	test negative	total
disease	3	1	4
no disease	9	87	96
total	12	88	100

Example 2. Consider the xkcd comic below. Explain the joke to someone at the level of a MAT 130 student.



Course Outline

- Unit 1 (Chapters 1-5)
- Unit 2 (Chapters 6-8)
- Unit 3 (Chapters 9-11, 13-14)